

Fixed-Point Wordlength Allocation for ADMM Hardware Accelerators

Internal technical report — complete record including refuted hypotheses

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Abstract

We study fixed-point wordlength allocation for an ADMM solver implemented on FPGA fabric. The central observation is that a proximal operator annihilates quantisation error on its degenerate set: the fraction d of coordinates where the operator’s derivative vanishes contributes no error to the output, so the effective noise scales with $(1 - d)$ rather than the classical $q^2/12$ applied uniformly. Measured against bit-exact simulation the resulting model achieves ratios of 0.99 (L1) and 1.00 (Box); the classical model is wrong by 3–4× typically and by 900× at $d = 0.999$. We derive a range-feasibility bound on the penalty parameter, pre-registered and confirmed. On routed Artix-7 silicon, narrowing the main datapath from Q2.16 to Q2.9 reduces LUTs by 46.6%, raises $F_{\max}$ by 28.8%, and cuts dynamic power from 11 mW to 7 mW, giving $184 \rightarrow 117$ nJ per solve. We report a negative result: per-operator asymmetric lane allocation costs 4.3% area at both main widths with no power or frequency compensation. All hardware figures are measured — SAIF-based power, swept $F_{\max}$, gate-level bit-exact validation — not estimated. We state explicitly where the model does not hold: it predicts ensemble-mean behaviour and under-predicts per-instance error by up to 7.6 bits.

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1 Introduction and scope

An accelerator for a proximal splitting method must choose a fixed-point format for every signal in its datapath. Too few fraction bits and the solver returns a wrong answer; too many and the fabric cost, critical path and switching energy all grow. The classical treatment models every quantisation as an independent additive noise source of power $q^2/12$ and propagates it through the algorithm. This report argues that treatment is substantially wrong for proximal operators, quantifies by how much, and validates the correction on silicon.

What is claimed. An error model in which the proximal operator’s degenerate set annihilates a measurable fraction of injected quantisation noise; a pre-registered range-feasibility bound; a measured demonstration that per-operator asymmetric allocation is net-negative in fabric; and a fully measured hardware characterisation of seven configurations.

What is not claimed. This is not the first error model for fixed-point ADMM — Jerez et al. have one. It is not a robustness bound. It does not supply a mechanism for every empirical relation reported. Section 8 lists the hypotheses that were tested and failed; Section 9 lists the scope limits.

2 Related work and precise positioning

2.1 The primary prior art

Jerez, Goulart, Richter, Constantinides, Kerrigan and Morari (arXiv:1303.1090) implement fixed-point ADMM on FPGA with an explicit round-off error analysis: a two-step error recurrence, Schur stability via Lyapunov argument, and a converging upper bound. They state the bound determines a priori the minimum number of bits for a given accuracy specification. **Any claim of a first error model for fixed-point ADMM is unavailable.**

The opening they leave is specific and is the basis of this work. In deriving their bound they observe that for the projection step no arithmetic error is introduced, and that the error *can only be reduced by multiplication with a diagonal matrix with entries componentwise in $[0, 1]$* . They then set that matrix to the identity to keep the bound conservative. That discarded diagonal is exactly the degenerate-set annihilation quantified here.

2.2 The second invitation

Kinsman and Nicolici (TCAD 30(9), 2011) allocate bit-widths for iterative computations by SMT, with conjugate gradient as their case study — the nearest published problem to ours. Their quantisation constraints, they write, *assume no correlation between quantisation error and value, when in reality correlation may exist. If the nature of the correlation is known, it can be captured into constraints.*

Their paper has sixteen citations in fifteen years and none extends it to iterative optimisation solvers; the descendants went toward range and overflow analysis rather than precision models for nonlinear operators.

2.3 The gap, confirmed by the incumbents

Ha and Sentieys state in both DATE 2020 and DATE 2023 that existing analytical techniques are *limited to modeling noise power metrics of Linear and Time-Invariant (LTI) systems (with some extensions)*. A proximal operator is not LTI. Two papers named the gap; a third confirms it was still open in 2023.

2.4 Comparisons that must not be made naively

Wu, Wallace, Mota et al. (J. Signal Process. Syst. 2022) report up to 67% LUT and 60% power reduction for mixed-precision ADMM. Those figures are *not* comparable to ours. Their baseline is FP32 floating point, so most of the gain is the float-to-fixed conversion itself; ours is Q2.16 fixed narrowed to Q2.9, with the float-to-fixed gain already banked in the baseline. Their power is estimated with the Xilinx Power Estimator; ours is measured from post-route switching activity. Their precision axis is temporal (per iteration); ours is spatial (per operator lane).

Usefully, their own Table 1 corroborates our negative result: for the ADMM solver in fixed point, consistent FXP24 uses 6775 LUTs while mixed FXP20→24 uses 6807 — non-uniform precision costing area, unremarked.

3 Error model

3.1 Degenerate-set annihilation

Let P_θ be a proximal operator with parameter θ , evaluated at fractional wordlength W with step $q = 2^{-W}$. Partition the input coordinates into the degenerate set D , where $\partial P/\partial v = 0$, and its complement.

Theorem 1 (Operator error decomposition).

$$\mathbb{E}\|e_P\|^2 = \frac{|D^c|}{n} \left[L^2 \frac{q^2}{12} + \sigma_{\theta, D^c}^2 \right] + \frac{|D|}{n} \sigma_{\theta, D}^2 \quad (1)$$

Writing $d = |D|/n$, the classical model overestimates by roughly $1/(1-d)$. On the degenerate set the operator’s output does not depend on the input, so injected perturbation is annihilated rather than propagated.

Validation. `model/error_decomp.py` measures the operator in isolation on a v -distribution harvested from float ADMM runs.

operator	model/measured	classical/measured	d range
L1 soft-threshold	0.99	3–4×	0.00–0.84
Box projection	1.00	3–4×	0.00–0.86
L2 ridge scaling	0.88	—	0

Table 1: Theorem 1 against bit-exact simulation. At $d = 0.999$ the classical model is wrong by 900×

[SCOPED: L2 gives 0.88, a 12% miss. L2 has $d = 0$, so the annihilation term is inactive and the discrepancy lies entirely in the θ term — the γ coefficient quantisation, which Corollary 1a states is not attenuated. The annihilation mechanism is validated; the coefficient-quantisation magnitude is not, to better than 12%. Quoting “0.99–1.00” without naming the operators would be selective.]

3.2 Range feasibility

Proposition 1 (Penalty lower bound). *Weight representability requires $\rho \geq 2^{-(I-1)}$ for I integer bits.*

This was registered before measurement. With $I = 2$ it predicts $\rho \geq 0.5$. Measured: $\max |M| = 2.116$ at $\rho = 0.125$ (infeasible in Q2.x), 1.117 at $\rho = 0.5$ (feasible). The pre-registration is the point and should be stated as such.

3.3 Loop-gain relation

Operator-level error does not determine solver-level error; the ADMM loop amplifies it. We fit

$$\log G = a \log \frac{1}{1 - \|M\|_2} + b R_{\text{op}} + c, \quad R_{\text{L1}} = d, \quad R_{\text{Box}} = d^2, \quad R_{\text{L2}} = \emptyset \quad (2)$$

where G is measured loop gain. Leave-one-out cross-validated error: 0.340 (L1), 0.234 (Box), 0.031 (L2) bits, all inside a 0.5-bit criterion.

The confound was tested. d correlates with conditioning at $r^2 = 0.94$, so the obvious objection is that it proxies conditioning where $\log(1/(1 - \|M\|_2))$ saturates. It does not: $\log(\text{cond})$ as the second regressor gives LOO 1.237 for L1, *worse than baseline*, and contributes nothing alongside d .

[SCOPED: This relation is empirical. No mechanism is claimed — see Section 8.2. The coefficient signs differ between L1 (+20.6) and Box (−20.4) because d moves in opposite directions with conditioning in the two operators ($r = +0.969$ and -0.975), not because two mechanisms are at work. Box prefers d^2 over d (paired LOO, $t = 2.56$, $p < 0.05$), so the operators do not share a functional form; several candidates were tried before this one, so the exponent is fitted, not derived.)]

3.4 Ordering of lane precision needs

Which operator needs the most lane precision is not fixed. Continuous W^* over a conditioning sweep shows the Box–L2 gap moving from -0.33 bits at $\kappa(H^\top H) = 2$ to $+2.33$ at 1000, crossing between 10 and 20.

The operator coefficient alone does not predict this: it gets the sign wrong in 6 of 8 resolvable Box–L2 cases and all 14 L1–L2 cases, missing by 1.2–1.5 bits. Multiplying by the loop resolvent $R = \|(I - T)^{-1}\|$, computed from M with no fitted parameter,

$$\Delta W_{ab} = \frac{1}{2} \log_2 \frac{A_a R_a}{A_b R_b}, \quad (3)$$

gives 0.288 bits and 8/8 signs for Box–L2 and places the crossover near 20. L2’s contraction pins its resolvent ($2.1 \rightarrow 3.2$) while L1 and Box grow ($2.7 \rightarrow 200$), so the contractive lane becomes relatively cheaper as the problem hardens.

[SCOPED: L1–L2 is 0.621 bits and misses the two smallest gaps in sign. L1–Box is under-predicted by 0.4–0.6 bits and the predictor is blind to κ ; the omitted κ -quantisation term of Corollary 1a is the likely cause and is untested. The exponents fitted in the loop-gain model make Box–L2 *worse* (1.05 bits, 4/8): the zero-parameter resolvent generalises across problem generators, the fitted model does not.]

4 Fabric cost model

Theorem 2 (Lane area). $A_{\text{lane}}(W) = bW + c$, with $b \approx 19.6$ LUT/bit.

[REFUTED: The originally proposed quadratic form $A(W) = aW^2 + bW + c$ fails its held-out test.] Fitting on the three no-cast points (lane width = F_{MAIN} : $16 \rightarrow 269.9$, $10 \rightarrow 159.0$, $9 \rightarrow 129.0$ LUT/lane) and predicting lane 8 (actual 122.0):

The fitted a is negative (-1.646): the measured slope *falls* from 30 LUT/bit ($9 \rightarrow 10$) to 18.5 ($10 \rightarrow 16$). A fabric-multiplier area law must be convex.

form	fit	predicts $W=8$	error
quadratic	exact (3 pts, 3 params)	95.7	−21.6%
linear	3.0% max residual	114.3	−6.3%

Resolution. The 166 vs 35 LUT measurement supporting a quadratic law was taken on the γ multiplier *in isolation*, where it does hold (4.74 measured against 4 predicted). At whole-lane granularity the multiplier is a small share — L1 and Box contain no multiplier at all — so casts, comparisons and saturating adds dominate and the total is linear. The W^2 term is not refuted; it is undetectable at lane granularity on this design. Both laws may be quoted provided the granularity is stated.

Cast cost, measured directly. Two builds share lane width 8 but differ in cast depth: 122.0 LUT/lane at $F_{\text{MAIN}} = 10$ against 127.5 at $F_{\text{MAIN}} = 16$, giving $c_{\text{cast}} \approx 0.9$ LUT/bit/lane.

5 Profitability of asymmetric allocation — a negative result

Theorem 3 (Profitability). *Asymmetric lane allocation is profitable only if area saved exceeds cast overhead. With lane area linear, saving is $b\Delta W$ — linear and small — so cast overhead dominates and asymmetry is net-negative.*

Note this derivation now rests on the linear law, which survived its falsifier, rather than on the rejected quadratic one.

matched pair (same F_{MAIN} , one lane narrowed)	LUT	F_{max}
asym_derived vs uniform10 ($F_{\text{MAIN}}=16$)	+4.3%	−5.5%
main9_asym vs main9_uniform ($F_{\text{MAIN}}=9$)	+4.3%	+1.3%

The area penalty is +4.3% at both widths, identical to one decimal place — a reproducible cost rather than two coincidentally similar numbers. Dynamic power is identical across three measured pairs. The F_{max} effect is inconsistent in sign and the +1.3% is one binary-search step, so no frequency claim is made in either direction.

Statement: per-operator asymmetry costs 4.3% area at both main-datapath widths, buys no measurable dynamic power, and has no consistent frequency effect.

6 Hardware results

6.1 Method

Artix-7 XC7A35T, out-of-context synthesis. Power from post-route gate-level simulation writing SAIF, consumed by `report_power`; every run Confidence High with 72–75% of nets matched, and bit-exact against the golden model in the same run that produced the activity. F_{max} by binary search on the clock constraint, seven full implementations per build, reported as 1000/(tightest period that met timing) — never 1000/(period − slack).

6.2 All seven builds

6.3 Headline

uniform18 → main9_uniform: LUT −46.6%, F_{max} +28.8% (bracketed +27.6% to +30.4% by the search resolution), dynamic power 11 → 7 mW, energy per solve 184 → 117 nJ.

build	F_{MAIN}	lanes	LUT	F_{max} (MHz)	dyn (mW)	total (mW)
uniform18	16	16/16/16	6308	66.3	11	80
uniform10	16	8/8/8	4439	67.6	10	78
asym_derived	16	8/7/8	4631	63.9	11	79
main10_uniform	10	10/10/10	4081	79.7	7	76
main10_lane8	10	8/8/8	3665	68.9	7	75
main9_uniform	9	9/9/9	3366	85.4	7	76
main9_asym	9	9/8/9	3510	86.5	7	76

Report absolutes, not the percentage. Power reports round to 1 mW, so 7 and 11 carry ± 0.5 mW and the honest bound on the reduction is 29–44%. Energy per solve is the better figure: dynamic energy per cycle is CV^2 and therefore frequency-independent, so -36% holds regardless of the operating clock a reader imagines.

Static power dominates. Device static is 68 mW of the 76–80 mW total and is identical across builds. Total on-chip power falls only 5.0%. **Any power claim must say *dynamic* explicitly or it is wrong by a factor of seven.**

6.4 Throughput

Cycles = $26K + 3$ exactly, across all seven builds, four main widths, symmetric and asymmetric. Cycles per iteration are invariant to every width parameter, so all performance gain comes from F_{max} .

[SCOPED: Iteration counts are *not* comparable across widths. With $\varepsilon = 0$ the early exit fires when the residual quantises to exactly zero, which happens sooner at coarser precision — stagnation, not convergence. main9 needs 15/19/12 iterations against uniform18’s 28/32/22; computing solutions per second from each build’s own iteration count bakes in a spurious $1.8\times$. Hold iterations fixed.]

6.5 Two critical-path regimes

Worst-path endpoints split the builds into two families with no overlap:

build	worst path endpoint	levels	F_{max}
uniform18, uniform10, asym_derived, main10_lane8	prox lane z.out_reg	21–24	63.9–68.9
main10_uniform, main9_uniform, main9_asym	converged_reg	16–17	79.7–86.5

$F_{\text{max}} = 1 / \max(\text{prox path, convergence-test path})$. Narrowing a lane below F_{MAIN} inserts a narrowing cast into the prox path; when that lifts it above the convergence-test path the build drops into the slow regime. main10_lane8 and main10_uniform isolate this to one variable: same F_{MAIN} , same main datapath, different lane width, different binding path, 11 MHz apart.

Design rule: narrowing the main datapath raises F_{max} only while the convergence-test path binds. Once a lane cast pushes the prox path above it, further narrowing buys area and nothing else.

Within-family spread is $\approx 8\%$ and is implementation variance. The family names the primal or dual residual interchangeably across builds; both feed **converged_reg** through the same comparison logic, so this is the *convergence-test* path and must not be described as a specific residual.

7 Measurement discipline

Three claims were withdrawn when vectorless estimates were replaced by measurement, and they are recorded because the near-miss is instructive.

1. Vectorless power was $\approx 10\times$ too high (112 vs 11 mW; 78 vs 7 mW). The ratio survived roughly intact by luck.
2. **[REFUTED: “Asymmetric pruning cuts dynamic power 18.8%”]** — measured, both configurations give 11 mW. The difference was an artefact of the estimator.
3. $F_{\max}$ of 71.1 MHz was $1000/(20\text{ ns} - \text{slack})$ at a deliberately loose constraint. Swept properly it is 85.4 MHz.

The near-miss. Comparing a vectorless `uniform18` (112 mW) against a measured `main9` (7 mW) would have produced a fabricated 94% power reduction. It was caught only because an object-count guard in the SAIF flow printed zero. Every number in Section 6 is measured, and the closest competing work estimates its power with a spreadsheet tool.

8 Hypotheses tested and refuted

Recorded because each would have read plausibly in a submission draft.

8.1 Composition (Theorem 5 as originally planned)

[REFUTED: 2.509 bits mean error against a 0.5-bit criterion.] The original theory placed contraction at the operator. The falsifier showed it is not there; contraction lives at the loop. The resolvent saturates at 3.18 with $g = 0.667$ against divergence to 47,669 at $g = 1$. This negative result produced the loop-gain reframing and was predicted in advance as the claim most likely to fail.

8.2 Active-set churn

The L1/Box coefficient sign difference invited a mechanism: L1’s degenerate set churns (entries crossing the soft threshold) while Box’s freezes (entries pinned to a clip bound). Churn was measured directly as the fraction of coordinates changing degenerate-set membership per iteration.

[REFUTED: Two of three predictions fail.] L1 churn averages 0.0011; Box averages 0.0028 — Box churns *more*, and both are $\approx 0.1\%$, i.e. both essentially frozen. Churn also fits worse than d (LOO 0.781 against 0.340 for L1). No mechanism is available and none is claimed.

8.3 F_{MAIN} dominance

An earlier claim that narrowing the main datapath dominates lane width on all three axes was carried by a confound: `uniform10` and `main10_uniform` differ in both parameters. The control build `main10_lane8` was added and **[REFUTED: the frequency half fails]** — holding lanes at 8, narrowing F_{MAIN} from 16 to 10 buys +1.9% $F_{\max}$, while *widening* lanes to match F_{MAIN} buys +15.7%. On area both axes are real and lanes are better per bit (-5.1% /bit against -2.9% /bit). The correct account is the two-regime mechanism of Section 6.5.

9 Scope limits

9.1 The model predicts ensemble means, not instances

`model/adversarial.py` attacks the fitted loop-gain model using Kinsman and Nicolici’s own construction — right-hand sides aligned across the eigenvectors of the system matrix — plus a greedy search maximising model error. Errors are reported signed, because the sign is the whole story: positive means the model *under*-predicts error, so wordlengths chosen from it would be too few.

operator	worst under-prediction, random rhs	adversarial
L1	+7.58 bits	+11.43 bits
Box	+2.16 bits	+3.43 bits
L2	+0.60 bits	+2.25 bits

[SCOPED: Random instances already break the 0.5-bit criterion. The cause is not an exotic corner: `loop_gain.py` fits 20-trial *ensemble averages* at each conditioning point, so the reported figures (0.834 / 0.399 / 0.031 bits) are ensemble-level quantities. A per-instance refit gives 0.979 / 0.611 / 0.339 bits before any attack.]

Consequence. The loop-gain relation is a design-space exploration tool predicting mean loop gain for an ensemble at fixed conditioning. It is *not* a per-instance safety guarantee. Every bit figure must be labelled “ensemble mean over N trials at fixed conditioning”.

What this does not invalidate. The Q2.9 design point was chosen with model guidance and then verified bit-exact against the golden model on routed silicon. The model proposes; measurement disposes. That division of labour is the methodology and should be stated, not hidden.

Why not a bound instead. Converting Theorem 1 into a guaranteed bound requires the Lyapunov or interval machinery that Jerez et al. and Kinsman and Nicolici already possess. Their bounds are conservative by construction — hundreds of bits for conjugate gradient, reducing to roughly single-to-double precision only after heavy input-space restriction — which is the deficiency a predictive model exists to address. Competing there means losing on their ground with their tools. [TODO: Quote a per-instance margin for wordlength selection, or state that per-instance safety requires the prior art’s bounds.]

9.2 Other limits

- Absolute LUT counts depend on the timing constraint — `uniform18` is 4673 at 20 ns and 6308 at 15.079 ns. The ratio survives (−44.9% against −46.6%) but tables at different constraints must never be mixed.
- Power reports round to 1 mW; this is the binding constraint on every power claim, including the apparent saturation at $F_{\text{MAIN}} = 10$.
- $N = 8$, one problem family, one device. No claim is made about scaling.
- The 0.5-bit criterion is looser than Li et al.’s < 1 dB (≈ 0.17 bits). [TODO: Justify the criterion rather than asserting it.]

10 Reproducibility

No number in this report exists without a script that produces it.

artefact	produced by
Theorem 1 validation	model/error_decomp.py
Loop-gain fit	model/loop_gain.py, loop_gain_l1.py
Churn refutation	model/box_form.py
Composition refutation	model/generalize.py
Adversarial scope test	model/adversarial.py
Golden vectors	model/gen_vectors.cfg.py
Builds	syn/run_ooc.tcl
SAIF power	syn/saif_power.bat, saif_log.tcl
$F_{\max}$ sweep	syn/fmax_sweep.tcl
Critical-path location	syn/critpath.tcl
Literature pass	tools/lit_sweep.py

11 Outstanding

- Board bring-up and measured wall power (Section 11 of the plan).
- Relabel all loop-gain figures as ensemble means.
- Justify the 0.5-bit criterion.
- Answer the per-instance margin question.